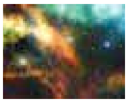
Greenland to lose ice
Greenland could lose 4.5% of its ice and add 13' of sea level by the end of this century. **https://dgit.in/july19-30**



Galaxy clusters' first kiss
Astronomers found that two large clusters of galaxies are just about to collide. **https://dgit.in/july19-31**

Professor Roy Martin, the researcher who serendipitously stumbled across the mounds while studying bees in the area, says, "It's incredible that, in this day and age, you can find an 'unknown' biological wonder of this sheer size and age still existing, with the occupants still present". While there are known examples of other massive termite mounds, the colonies associated with them are no longer active. The mounds in brazil are still continuing to grow, and are unlike other mounds. The termites do not live in the mounds, and these are not nests. Instead, the mounds are piles of waste material, ejected from the ground because of the excavation of a network of underground tunnels. The tunnels allow the termites to forage for food over large areas, without being exposed to predators. The food is scarce in the area, and even a leaf drying up and falling to the ground is a rare event. To find this rare food, the termites have to spread out over a large area.

Each of the mounds are about 2.4 meters in height and about 9 meters in diameter. All the mounds are roughly cone shaped. Each mound has about 50 cubic meters of soil. To date the mounds, the researchers collected soil samples from the centers of 11 mounds, and shipped them to a specialised laboratory at the University of Nebraska-Lincoln, that could date when the grains of sand were last exposed to sunlight. The youngest sample was dated to 680 years ago, while the oldest was dated to 3,820 years ago. The dates are comparable to the oldest known termite mounds elsewhere in the world. As only a few of the over 200 million mounds were sampled, it could mean that there are mounds out there that are even older. The findings indicate that a large number of mounds were built over different periods of time. The researchers plan to date samples from more mounds to get a better idea of the range of dates. The method could also allow dating of particular portions of mounds, to understand when they were created. Multiple samples of a same mound can allow the researchers to find out how the mounds grew over time. Apart from the dating, there is also the spacing between the mounds. This is one of the simplest ways to understand patterns in nature. Eun-Hye "Enki" Yoo from the University of Buffalo was drafted to act as the geographer in the scientific team, while she was visiting Brazil for a holiday. Yoo used satellite imagery to measure the distances between the mounds. "As a geographer, it's a routine task to examine the spatial pattern of any subject matter, whether they are trees, human mobility, cities or even termite mounds. We find so many hidden stories by carefully exploring their spatial patterns. The identification of the termite mounds' unique pattern was helpful to tease out some hypotheses and to explain why they were built in a certain way. For example, if they happened to be clustered, we would have been interested in explaining why they were clustered — for what? Food or water sources?" The analysis showed that resources were not a factor in the regular spacing of the mounds. Instead, they were an efficient way to space out the dumps, which are essentially piles of garbage for the termites. This optimisation is a process that took place over millennia. The real magic is below the mounds, in the network of tunnels. The spacing of the mounds allow for the termites to safely navigate from any location within the colony, to a semi regular supply of food. Yoo wants to take more precise measurements of all the 200 million mounds, and also wants to find out if climate change affects the spacing.

One of the easy ways to inspect the mounds, was to observe the ones that had been bisected by the construction of roads. The mounds did not have any complex internal structures, and were just amorphous piles. There is just a single tunnel from the center of the mound to the subterranean termite labyrinth. Between the mounds are temporary tunnels that are opened during the night. Each night, small bands of between 10 to 50 soldier termites crawl out and forage for dried leaves. Once the food is found, they crawl back in, closing the temporary tunnels behind them. The harvested leaves are stored in narrow galleries. Despite searching for it, the researchers were unable to locate a queen termite or even a royal chamber. This shows that the network of tunnels is effective at protecting the location of the termite queen. As the tunnels are closed up at night, and not left open

Credit: Eun-Hye "Enki" Yoo

The termite mounds as seen on Google Earth

to the environment, the researchers could conclude that the tunnel network did not serve the function of ventilation. Another species that also lives in arid regions, the Heterocephalus or the naked mole rat (do not Google search the image of the creature unless you want to get nightmares), also builds extensive subterranean tunnel networks to make foraging for food easier. The naked mole rat also lives in colonies, with individuals taking on roles of a queen and workers.

The discovery of the termite mounds in Brazil is a recent one, and everything we know about them is only the result of preliminary studies. The researchers involved with the initial paper plan to investigate the mounds further. We are just beginning to understand this impressive bioengineering feat that has been going on for thousands of years. d